Model Program Book

COMMUNITY SERVICE PROJECT





Designed & Developed by

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH)

WATER POLLUTION

-- turn our waters blue, not brown

Community Service Project report submitted at the end of 2024-2025 Academic Year in partial fulfillment of the requirement for the award of the degree of :-

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING – DS



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING RAGHU ENGINEERING COLLEGE

(Approved by AICTE, New Delhi, Accredited by NBA (CSE, ECE, CIVIL&MECH)

& Re Accredited by NAAC with 'A+' grade & Permanently Affiliated to JNTU-GV Vizianagaram) Dakamarri, Bheemunipatnam Mandal, Visakhapatnam Dist. – 531162 (A.P.)

2024- 2025

Program Book for
Community Internship

Name of the Project :

WATER POLLUTION

Name of the students :

•VADAMA SURENDRA

(22981A44I7)

•SHAIK KAJAMOHIDDIN

(22981A44G9)

•RUTTALA DESHIK SAI

(22981A44G2)

• RAJAHMUNDRY.JYOTHI BABU

(22981A44F2)

Name &Address of the Community/Habitation:

WATER POLLUTION

&

Ramakrishna Beach, Visakhapatnam, Andhra Pradesh

Certificate from
Official of the Community

This is to certify that _____ *(Name of the Community Service Volunteer)* Reg. No _____ of _____(Name of the College) underwent community service in _____*(Name of the Community)* from _____ to _____ The overall performance of the Community Service Volunteer during his/her community service is found to be _____ *(Satisfactory/Good).*

Authorized Signatory

with Date and Seal

Student's Declaration

We Vadama.Surendra, Shaik.Kajamohiddin ,Ruttala Deshiksai,R.JyothiBabu, Regd.no – 22981A44I7 ,22981A44G9,22981A44F2, 22981A44G2 of the Department of CSD Raghu

Engineering College,College does here by deciare that I have completed the mandatory community internship from **/**/**** to **/**/**** in ***Water Pollution.***

Student Signature with date **Signature of Mentor**

Signature of Community Internship coordinator(department)

Signature of Community Internship coordinator(college)

Signature of Head of the Department

DEPARTMENT VISION AND MISSION DEPARTMENT PEOs

Vision of the Department:

To generate competent professionals to become part of the industry and research organizations at the national and international levels.

Mission of the Department:

M1: To impart high quality professional training in undergraduate level with emphasis on Basic principles of Computer Science and Engineering and to foster leading edge research in the fast changing field.

M2: To inculcate professional behavior, strong ethical values, innovative research capabilities and leadership abilities in the young minds so as to work with a commitment.

Program Educational Objectives (PEOs):

PEO1: To produce graduates who have strong foundation in mathematics, science, Engineering fundamentals, laboratory, and work-based experiences to formulate and solve engineering problems in computer science engineering domains and shall have proficiency in implementation software tools and languages.

PEO2: To progressively impart training to the students for success in various engineering positions within the core areas in computer science engineering, computational or adapting themselves to latest trends by learning themselves.

PEO3: To produce graduates having the ability to pursue advanced higher studies and research. To have professional and communication skills to function as leaders and members of multidisciplinary teams in engineering and other industries with strong work ethics,organizational skills, teamwork and understand the importance of being a thorough professional.

POs & PSOs

Program Outcomes (POs):

1.Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to solve complex engineering problems.

2.Problem analysis: Identity, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3.Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and the cultural, societal, and environmental concerns.

4.Conduct investigations of complex problems: Use research-based knowledge and

research methods, including design of experiments, analysis, interpretation of data, and synthesis of the information to provide valid conclusions.

5.Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex engineering activities with an understanding of the limitations.

6.The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7.Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.

8.Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of the engineering practice.

9.Individual and teamwork: Function effectively as an individual and as a member or leader in diverse teams and multidisciplinary settings.

10.Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11.Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work as a member and leader in a team, to manage projects and in multidisciplinary environments.

Program Specific Outcomes (PSOs):

PSO1:

Apply the concepts and techniques of the Computer Science & Engineering branch and the mathematical foundations in the significant domains to address the complex engineering problems concerning environmental, safety, economics, culture, and society.

PSO2:

Employ emerging computer languages and platforms in developing innovative career prospects as an entrepreneur with leadership, ethical, and communication skills with a zest to pursue higher studies in the field of Computer Science & Engineering.

PSO3:

Apply the managerial, interdisciplinary skill set, and domain-specific tools in working system processes to implement and deploy a quality-based software product to meet evolving needs.

COURSE OBJECTIVES

(Community Internship)

To sensitize the students to the living conditions of the people who are around them. To help students to realize the stark realities of the society.

To bring about an attitudinal change in the students and help them to develop societal consciousness, sensibility, responsibility, and accountability.

To make students aware of their inner strength and help them to find new/out of box solutions to the social problems.

To make students socially responsible citizens who are sensitive to the needs of the disadvantaged sections.

To help students to initiate developmental activities in the community in coordination with public and government authorities.

To develop a holistic life perspective among the students by making them study culture, traditions, habits, lifestyles, resource utilization, wastage's and its management, social problems, public administration system and the roles and responsibilities of different persons across different social systems.

ACKNOWLEDGEMENTS

We express our sincere gratitude to our esteemed institute “Raghu Engineering College”, which has provided us an opportunity to fulfil the most cherished desire to reach our goal.

We take this opportunity with great pleasure to put on record our ineffable personal indebtedness to Mr. ********, Chairman of Raghu Engineering College for providing necessary departmental facilities.

We would like to thank the principal Dr. Ch Srinivasu, Administration and Management of “Raghu Engineering College”, for providing the requisite facilities to carry out the project on campus.

We sincerely thank Mr. *******, Head of Department, Computer Science and Engineering, Raghu Engineering College, for his kind support in the successful completion of this work.

We extend thanks to the people of Ramakrishna ,Visakhapatnam.

We sincerely express our deep sense of gratitude to******** Associate Professor, Department of Computer Science and Engineering, Raghu Engineering College, for his kindness and wisdom coupled with patience. It’s a great pleasure of ours to submit this project under his wing.

We extend thanks to the faculty members of the Computer Science Department for their value based imparting of theory and practical subjects which were used in the project. We are thankful to the non-teaching staff of the Department of Computer Science and Engineering, Raghu Engineering College, for their inexpressible support.

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Chapter-1: PROBLEM STATEMENT

Water pollution is the contamination of water bodies, usually as a result of the human activities, so that it negatively affects its uses, Water bodies include lakes, rivers, oceans, aquifers, reservoirs and groundwater. Water pollution results when contaminates mix with these water bodies. Contaminates can come from one of four main sources: sewage discharges, industrial activities, agricultural activities, and urban runoff including stormwater.

Water pollution is either surface water pollution or groundwater pollution. This form of pollution can lead to many problems, such as the degrading of aquatic ecosystems or spreading water bone diseases when people use polluted water for drinking or irrigation. Another problem is that water pollution reduces the ecosystem services that the water resource would otherwise provide.

Sources of water pollution are either point sources or non-point sources. Point sources have one identifiable cause, such as a storm drain, a waste water treatment plant or an oil spill. Non-point sources are more diffuse, such as agricultural runoff. Pollution is the result of the cumulative effect over time. Pollution may take the form of toxic substances (e.g.., oil, metals, plastics, industrial waste products) or the introduction of pathogenic organisms. Control of water pollution requires appropriate infrastructure and management plans as well as legislation. Technology solutions can include improving sanitation, sewage treatment, industrial waste water treatment, agricultural waste water treatment, erosion control, sediment control and control of urban runoff including storm water management.

Chapter-2: Executive Summary

After completion of choosing the town and finding the problem of people suffering about the water pollution we decided to create the awareness about the water pollution. We had shared the information about the greatness of the water and it uses. We had created the awareness about the water pollution by using the poster presentation, plug cards, and by designing some charts. We had got interacted with many people including the farmers to create the awareness about the water pollution. We had explained the causes of the water pollution by saying the following things such as

Not only is the agricultural sector the biggest consumer of global fresh water resources, with farming and livestock production using about 70% of the earths water supplies, but it's also a serious water polluter. Around the world, agriculture is the leading cause of water degradation. In the United States, agricultural pollution is the top source of contamination in rivers and streams, the sewage and waste water also causes the water pollution. Due to this 850 billion people are of untreated by waste water each year.

By the oil pollution also we are the getting the water polluted. As the oils are naturally released from under the ocean floor through fractures known as seeps. By explained all of this causes we had created awareness about the water pollution.



ACTIVITY LOG FOR THE FIRST WEEK

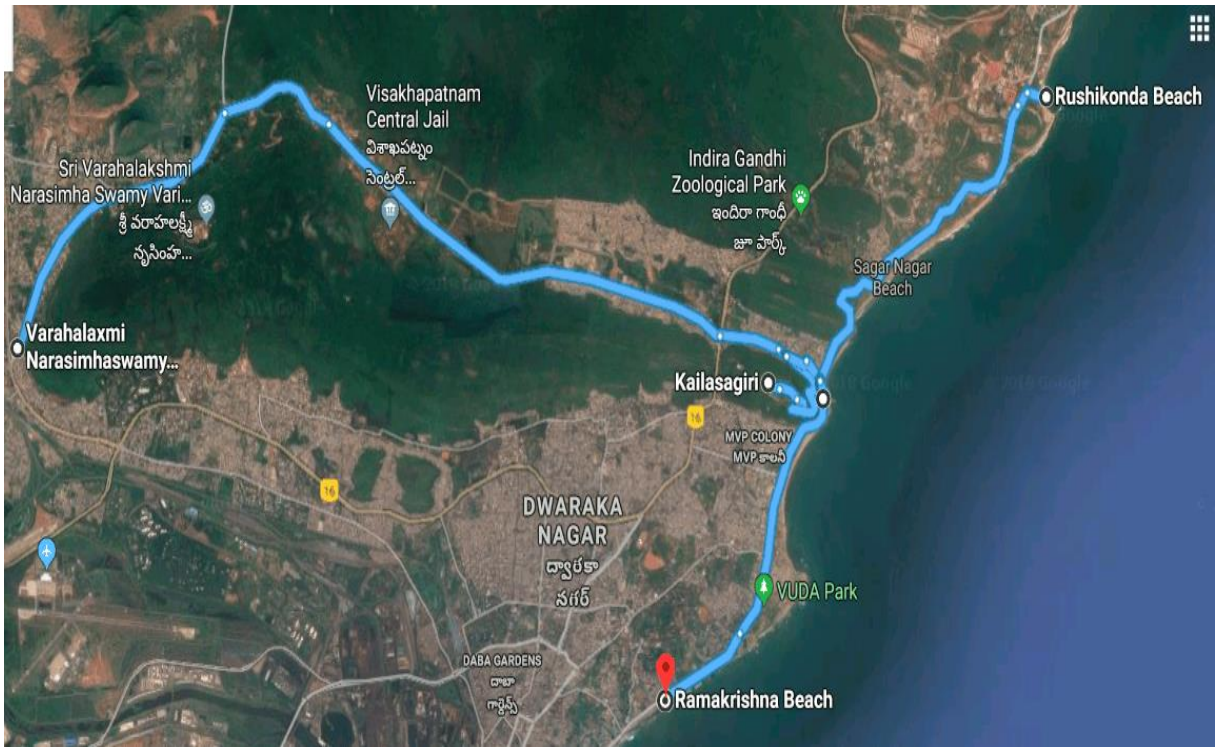
Day & Date	Brief description of the daily activity.	Learning Outcome	Mentor Signature
Day -1	Introduction of community service project. Community Survey: A community Survey is a method of collecting data from a filtered target audience to help you understand an issue particular to them. Taking guidelines from mentor about community service project.		
Day -2	We discussed the Needs and importance of community service project and analyzed the project.		
Day-3	We gathered the forms of community services and Selection of community service type (direct service). And finalized the place from where to start the project.		
Day - 4	We visited beach and water polluted areas and collected the information on it.		
Day - 5	Then we visited the ramakristana beach noticed some areas which are polluted.		
Day - 6	We meet the government workers and took permission to do this CSP project in this area or Site.		

Chapter-3: OVERVIEW OF THE COMMUNITY

We team members are discussed about place for our project and we decided to choose the town which is near to all of us so that we can flexibly visit the town whenever we decide to go there. After some time of discussion, we decided to visit all nearby town to find the problem, after starting the exploring our nearby towns and villages we found the problem in Ramakrishna beach. So that we choose the town Ramakrishna beach as our place for project.

We are a team of 4 and we have visited a field near Ramakrishna beach, Visakhapatnam. We went through the fields and then had some conversation with the lady working in the beach.

If you are one of those vizagites who's proud of our long coastline and unspoilt , pristine beach. RK Beach, also known as Ramakrishna Beach, is a popular beach in Visakhapatnam, Andhra Pradesh that's known for its natural beauty and scenic views The Ramakrishna beach is a beautiful spot for relaxation and pleasure is a natural beauty scene. There is an imposing Kali Temple near this beach. The aquarium here is an added attraction. The roaring sounds of sea gives enormous pleasure.



Fg.1- AREAL VIEW OF Ramakrishna beach, Visakhapatnam.

Chapter-4: COMMUNITY SERVICE PART

When we went to Ramakrishna beach to find a problem, we found that this town is full of Population but we found that the water resources are totally polluted so that we decided to create the awareness among the topics of causing and reducing the water pollution. The people who are using the polluted water will gets lots of health issues so we had selected this topic “WATER POLLUTION”.



The World Health Organisation (WHO) says that polluted water is water whose composition has been changed to the extent that it is unusable. In other words, it is toxic water that cannot be drunk or used for essential purposes like agriculture, and which also causes diseases like diarrhoea, cholera, dysentery, typhoid and poliomyelitis that kill more than 500,000 people worldwide every year.

The main water pollutants include bacteria, viruses, parasites, fertilisers, pesticides, pharmaceutical products, nitrates, phosphates, plastics, faecal waste and even radioactive substances. These substances do not always change the colour of the water, meaning that

they are often invisible pollutants. That's why small amounts of water and aquatic organisms are tested to determine water quality.



The effects of water pollution are far-reaching, impacting ecosystems, drinking water quality, and public health. Solutions include better waste management practices, stricter regulations, and increased public awareness about conservation. Reducing pollution at the source and promoting sustainable practices are crucial for preserving water quality for future generations.



Hotel waste can significantly contribute to pollution at beaches, impacting both the environment and local communities. Here are some key points about this issue:

1. **Wastewater Discharge**: Many hotels may discharge untreated or inadequately treated wastewater directly into nearby water bodies. This can introduce harmful bacteria, chemicals, and nutrients that degrade water quality.

2. **Solid Waste**: Hotels generate substantial amounts of solid waste, including plastics, food scraps, and other materials. If not properly managed, this waste can end up on beaches, harming wildlife and polluting the ecosystem.



Chapter-5: SURVEY METHODOLOGY

Methodology:

1. Physical Observation:

We found that this town contains totally polluted water over the area then we came to understand the situation in the locality and decided to create the awareness about the water pollution. We had decided to create the awareness on the industries, agricultural activities by using water, oil spills, and many other things.



2. Interviews:

It is an interesting thing that some people had shared their valuable information regarding water pollution reduce methods with us, they shared about the industries and the wastage dumping by the factories, and the problems that they are suffering due to the waste water. And also they shared the information about the drainage water that are directly entering into the water sources.

3. Data Collection:

Data collection is the means by which these types of data are acquired for multiple uses, including for flood warnings and other health and safety monitoring activities, weather prediction, engineering design, commercial and industrial applications, and scientific research.

Chapter-6: SUMMARY OF DATA COLLECTIONS & INTERVIEWS

SUMMARY OF DATA COLLECTIONS & INTERVIEWS

Summary of data collection:

1. We collected the brief data related to people with which we interacted.

2. We collected the information related to their details, their family members details, their occupation.

3. We also collected that whether they are aware of water pollution or not.

4. We collected about technical information means, whether they knew about mobile applications related to water pollution or not.

The below pictures are the survey sheets in which we included all the questionaries.


GPS Map Camera
Visakhapatnam, Andhra Pradesh, India
P869+CC7, Urban, Visakhapatnam, Andhra Pradesh 530003, India
Lat 17.711324°
Long 83.318539°
22/09/24 11:32 AM GMT +05:30
Google


GPS Map Camera
Visakhapatnam, Andhra Pradesh, India
P868+2J6, Visakhapatnam, Andhra Pradesh 530003, India
Lat 17.70969°
Long 83.316354°
22/09/24 11:07 AM GMT +05:30
Google





Interviews:

Some people shared their valuable information about the causes and reducing methods with us.

We met P. Rani garu, GVMC worker she shared his valuable informing about the one cause of the water pollution in our interview that the using and dumping the wastage into the lakes and rivers causing polluting the water due to that the aquatic animals are dying.



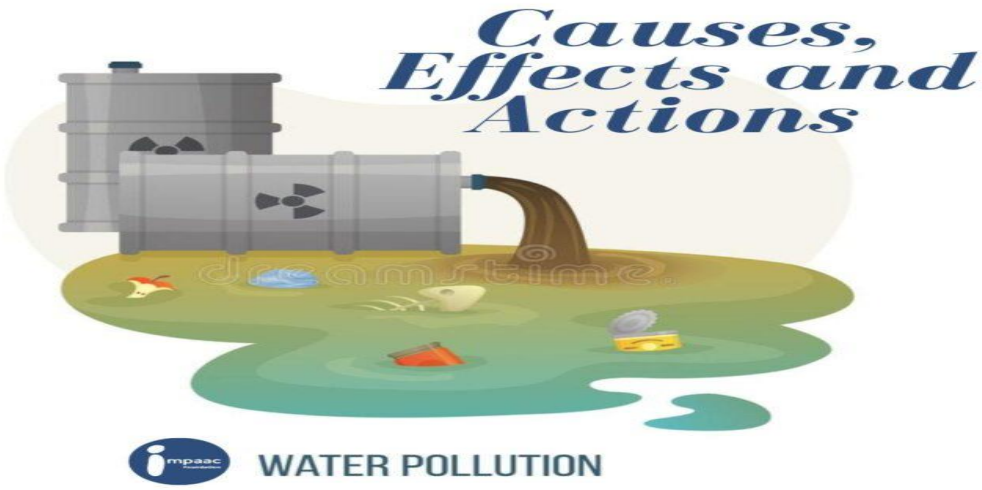


Photographs with information:

->Mixing of polluted water in ocean



->Causes, effects and actions



We had created our awareness by explaining all the things which we had learnt form the guide and the websites. In the town the water is polluted due to waste throwing into the water sources. So we created the awareness about the water pollution.



By dumping the wastage into the water sources the aquatic animals are suffering in the water to take breath and survive under water. It causes the reduce of the aquatic animals in the water sources.



All of them are responded positively and they said us that they will definitely control the water pollution. They promised to use only the natural bags suggested by us. We get huge positive responses from the people we enjoyed a lot by doing this project and also to interact with the people.

Chapter-7: OUTCOME OF THE PROJECT

Learning Outcomes:

•Positive impact on students' academic learning in view of the classroom to field and vice versa experience.

•Improves students' ability to apply what they have learned in "the real world".

•Positive impact on academic outcomes such as demonstrated complexity of understanding, problem analysis, problem-solving, critical thinking, and cognitive development.

•Improved ability to understand complexity and ambiguity.

Personal Outcomes:

•Greater sense of personal efficacy, personal identity, compassion, spiritual growth, and moral development.

•Greater interpersonal development, particularly the ability to work well with others, and build leadership and communication skills.

Social Outcomes:

•Learning service as a graduate attribute.

•Reduced stereotypes and greater inter-cultural understanding.

•Improved social responsibility and citizenship skills.

•Greater involvement in community service after graduation.

Relationship with the Institution Academic year:

•Stronger relationships with faculty.

•Greater satisfaction with college.

•Improved graduation rates.

Chapter-8: CONCLUSON &SCOPE FOR TECHNOLOGY TO ADDRESS THE IDENTIFIED PROBLEMS

Conclusion:

we decided to create the awareness about the water pollution. We had shared the information about the greatness of the water and it uses. We had created the awareness about the water pollution by using the poster presentation, plug cards, and by designing some charts. We had got interacted with many people including the farmers to create the awareness about the water pollution. We had explained the causes of the water pollution by saying the following things such as:

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★SAVE WATER SAVE LIFE★




Let's Stop
WATER
POLLUTION


SAVE WATER

Student self-Evaluation for the community service project

(Rating scale 1 is lowest and 5 is the Highest rank)

1) Oral communication	1	2	3	4	5
2) Written communication	1	2	3	4	5
3) Proactiveness	1	2	3	4	5
4) Interaction ability with community	1	2	3	4	5
5) Positive Attitude	1	2	3	4	5
6) Self-confidence	1	2	3	4	5
7) Ability to learn	1	2	3	4	5
8) Work Plan and organization	1	2	3	4	5
9) Professionalism	1	2	3	4	5
10) Creativity	1	2	3	4	5
11) Quality of work done	1	2	3	4	5
12) Time Management	1	2	3	4	5
13) Understanding the Community	1	2	3	4	5
14) Achievement of Desired Outcomes	1	2	3	4	5
15) Overall Performance	1	2	3	4	5

Date signature of the student Evaluation by the faculty mentor

Rating scale :1 is the lowest and 5 is Highest rank

1) Oral communication	1	2	3	4	5
2) Written communication	1	2	3	4	5
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Signature of the Faculty Mentor

Date: